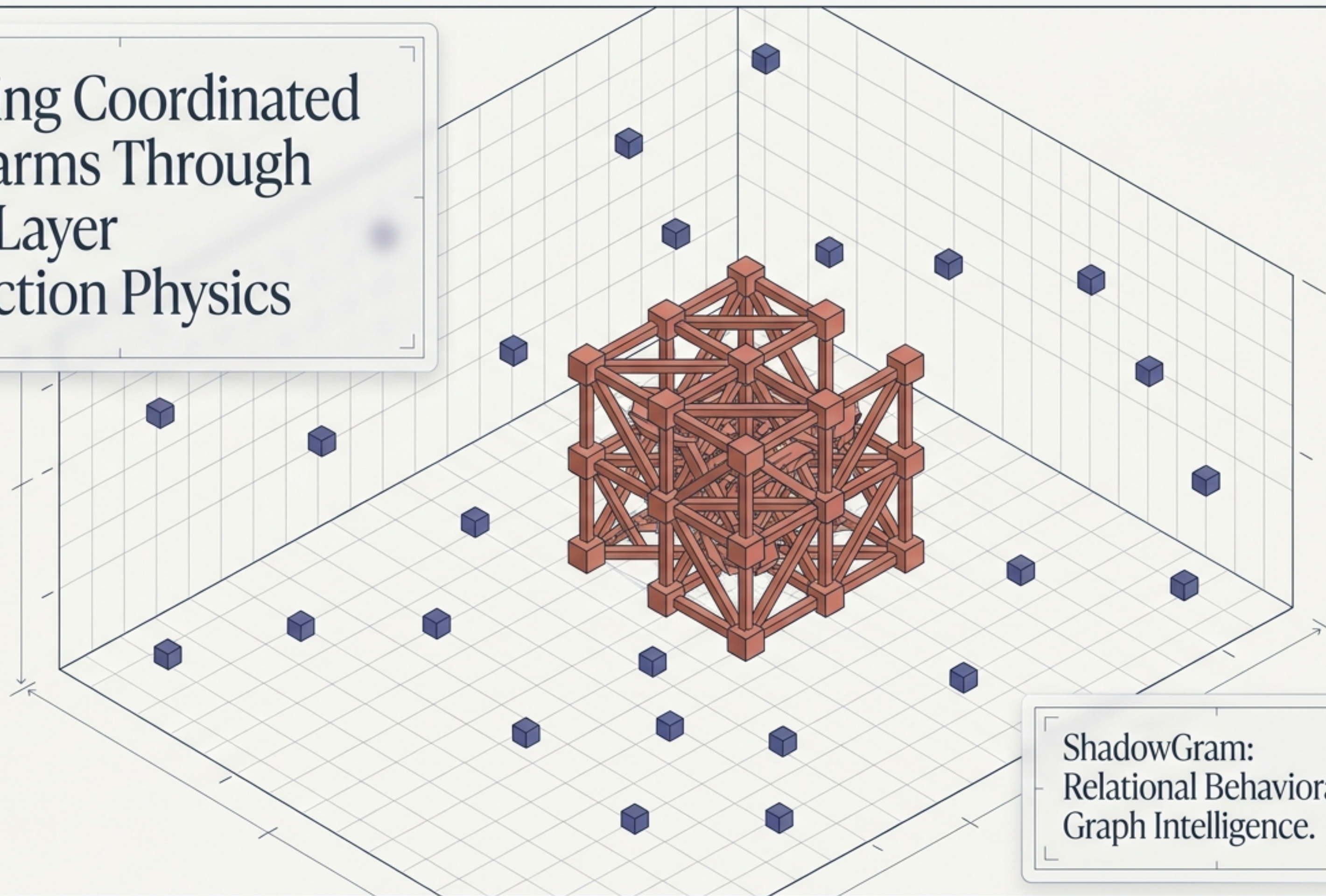


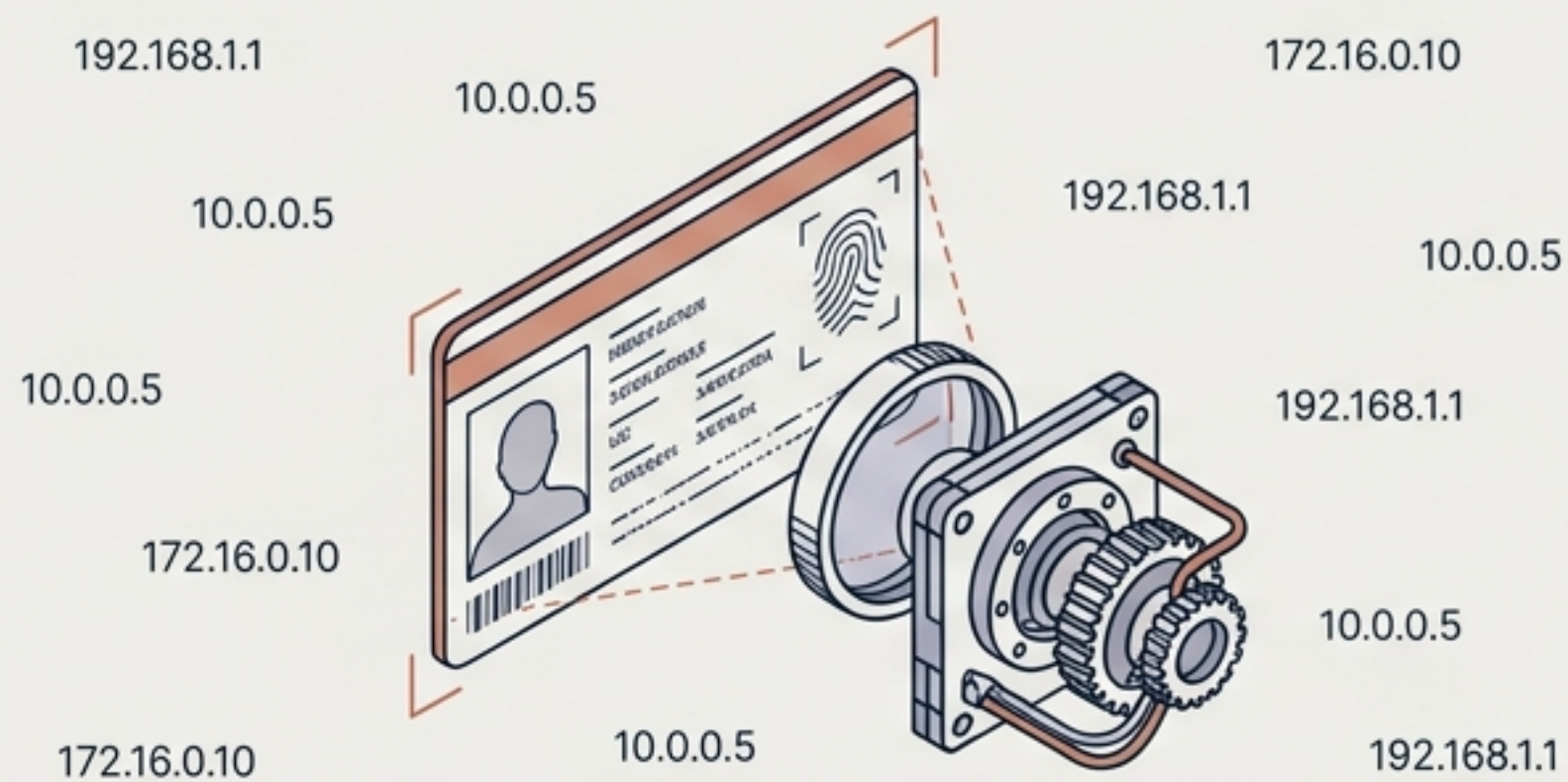
Catching Coordinated AI Swarms Through Multi-Layer Interaction Physics



ShadowGram:
Relational Behavioral
Graph Intelligence.

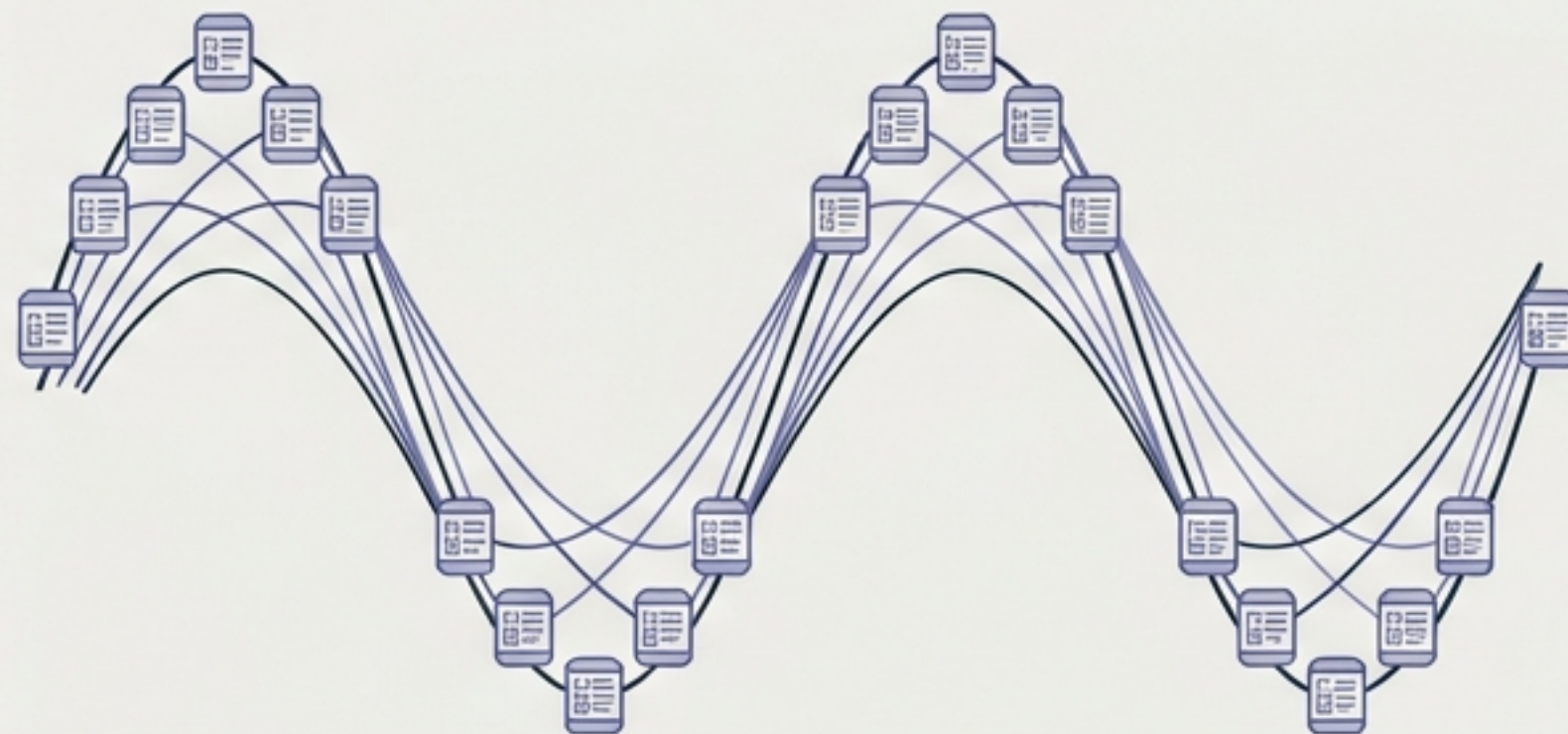
Moving from analyzing isolated accounts to illuminating behavioral bonds

Point-in-Time Isolation



Inspects isolated IPs, devices, and CAPTCHAs. Scammers easily randomize parameters, making every bot look 100% human in isolation.

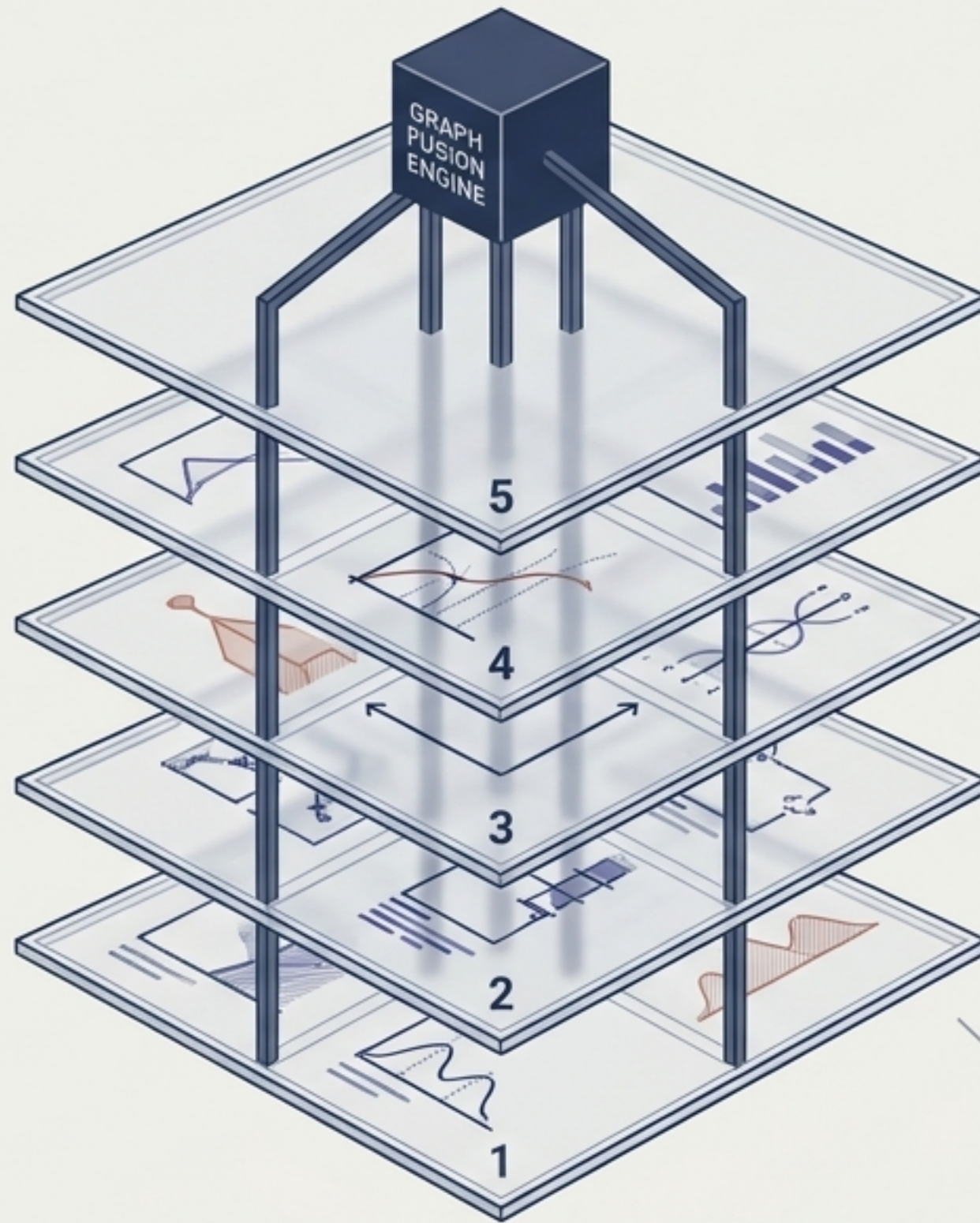
Relational Session Topology



Evaluates cross-session behavioral correlations in real time. Identifies the underlying automation physics binding the swarm together.

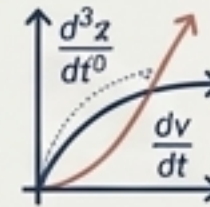
Even when bots use different names and IPs, the underlying **automation** forces them to move with identical mechanical physics.

Evading detection requires randomizing five behavioral layers simultaneously



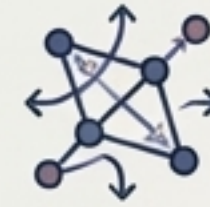
Environmental Entropy (S_{env}):

AudioContext hashes, WebGL canvas rendering, hardware signatures.



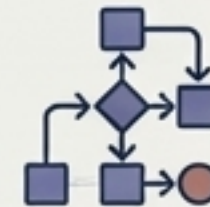
Kinetic Micro-Biometrics (S_{kin}):

Mouse trajectory curvature and jerk derivatives (d^3x/dt^3).



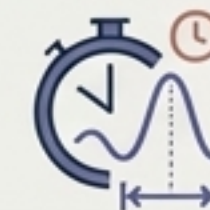
Semantic Embedding NLP (S_{sem}):

Cosine similarity in 384-dimensional dense vector space.



Navigation State Machine (S_{nav}):

Longest Common Subsequence (LCS) of DOM clicks and route transitions.



Temporal Rhythm (S_{time}):

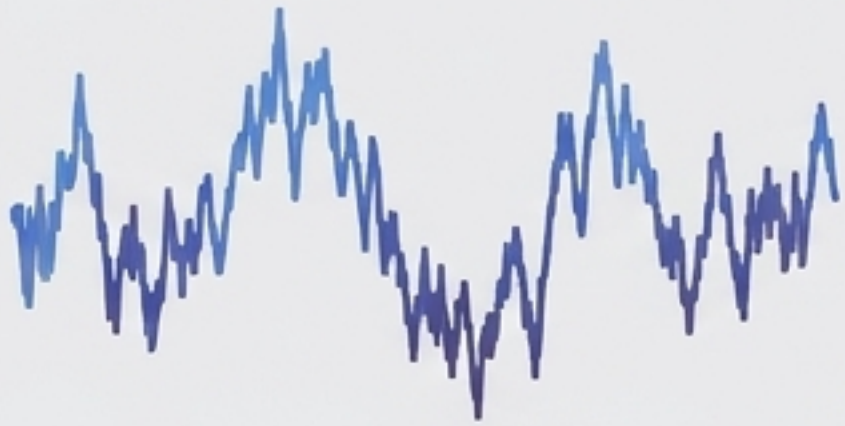
Arrival intervals (Δt), burst synchronization, sub-millisecond pacing.

ShadowGram calculates a composite similarity score across five independent layers of interaction physics.

Evaluating micro-muscle tremors versus mathematical perfection in under 10 milliseconds

Human User Pipeline

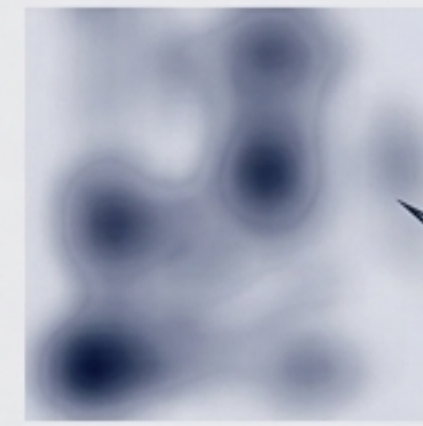
Cursor Trajectory



Continuous Acceleration Variance



128x128 soft, dispersed density spectrogram



A lightweight 2D-CNN
exported to ONNX Runtime
evaluates the spectrogram
image on local CPU: Output
 $P_{\text{synthetic}} \in [0.0, 1.0]$ in $<10\text{ms}$.

Bot Syndicate Pipeline

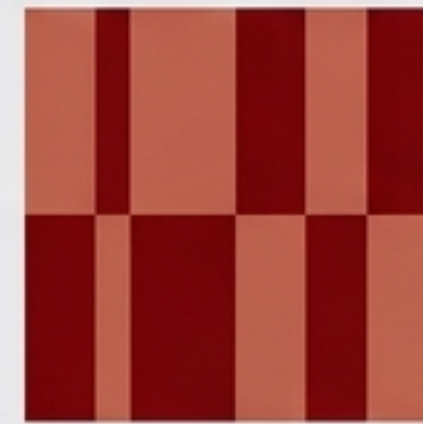
Cursor Trajectory



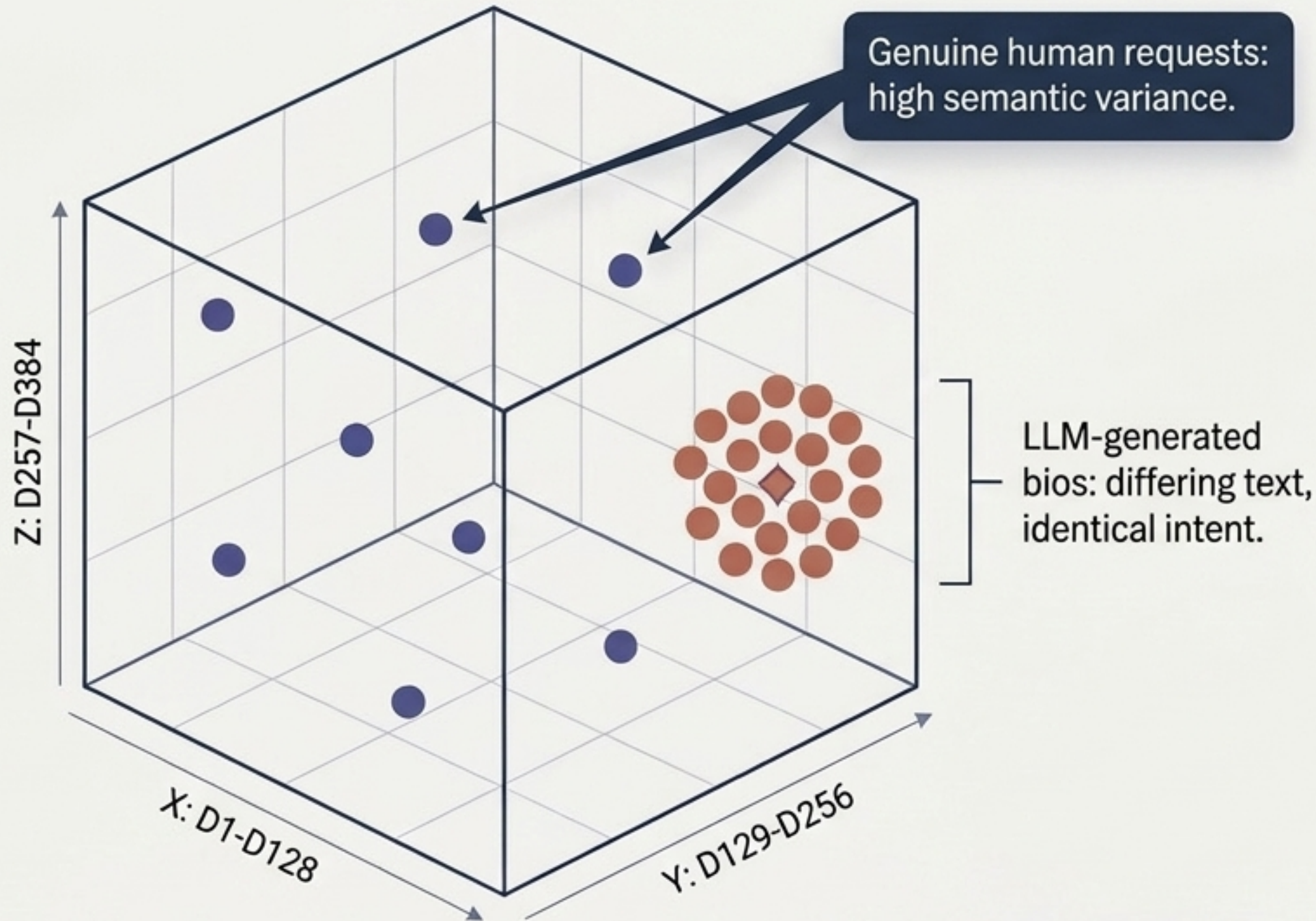
Stepped Acceleration ($d^3x/dt^3 \approx 0$)



128x128 harsh, rigid, spectrogram



High-dimensional semantic clustering catches LLM-generated variations instantly



Swarms use LLMs to generate unique loan reasons.

String matching fails, but semantic meaning remains identical.



Local CPU Model

(sentence-transformers/all-MiniLM-L6-v2)
detects

$$\text{Cosine}(\mathbf{A}, \mathbf{B}) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} \geq 0.88 \text{ in } < 80\text{ms.}$$

ShadowGram calculates a composite similarity score across five independent layers of interaction physics.

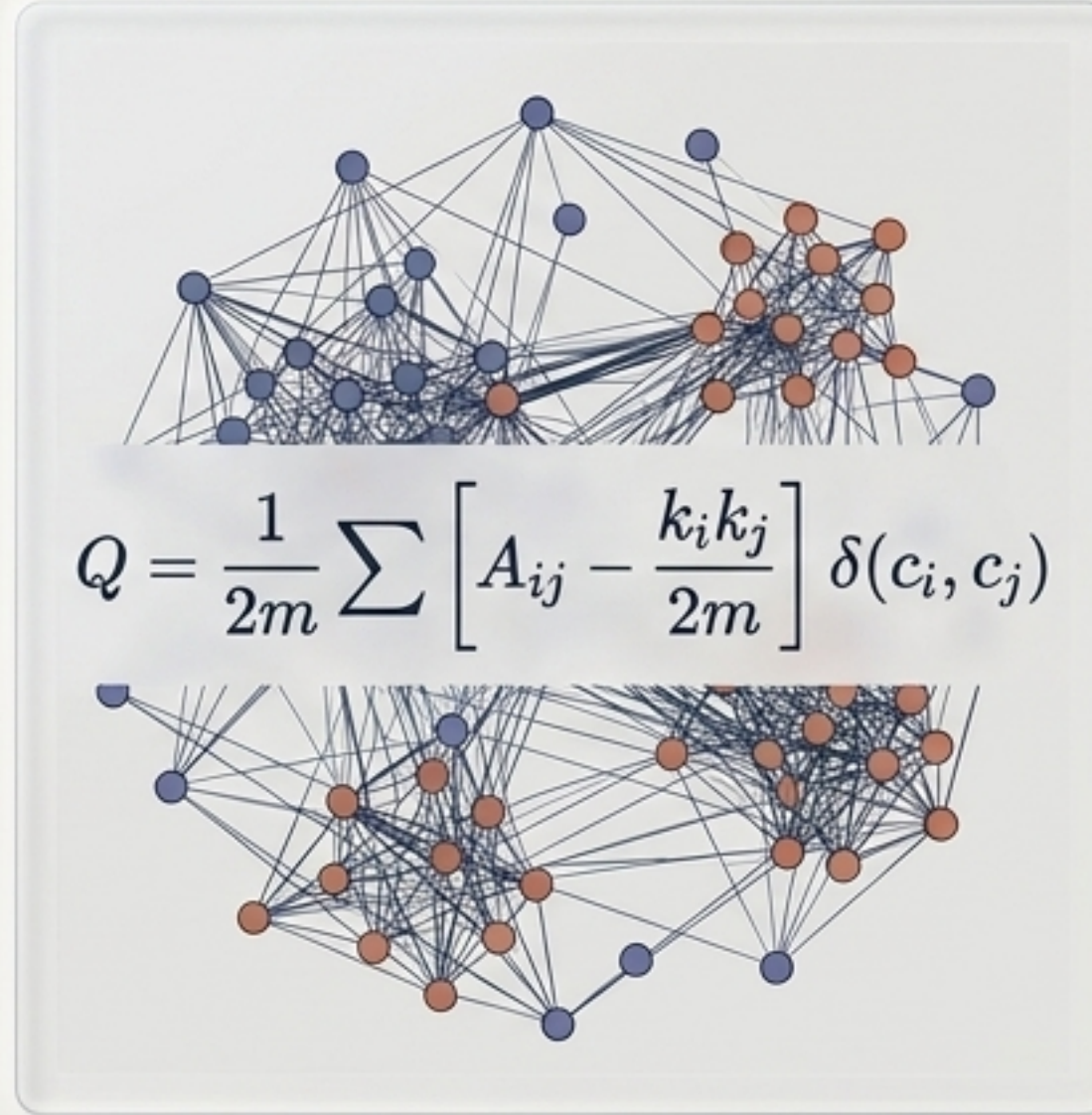
Mathematics draws the exact boundaries of the criminal syndicate automatically

1. Data Ingest



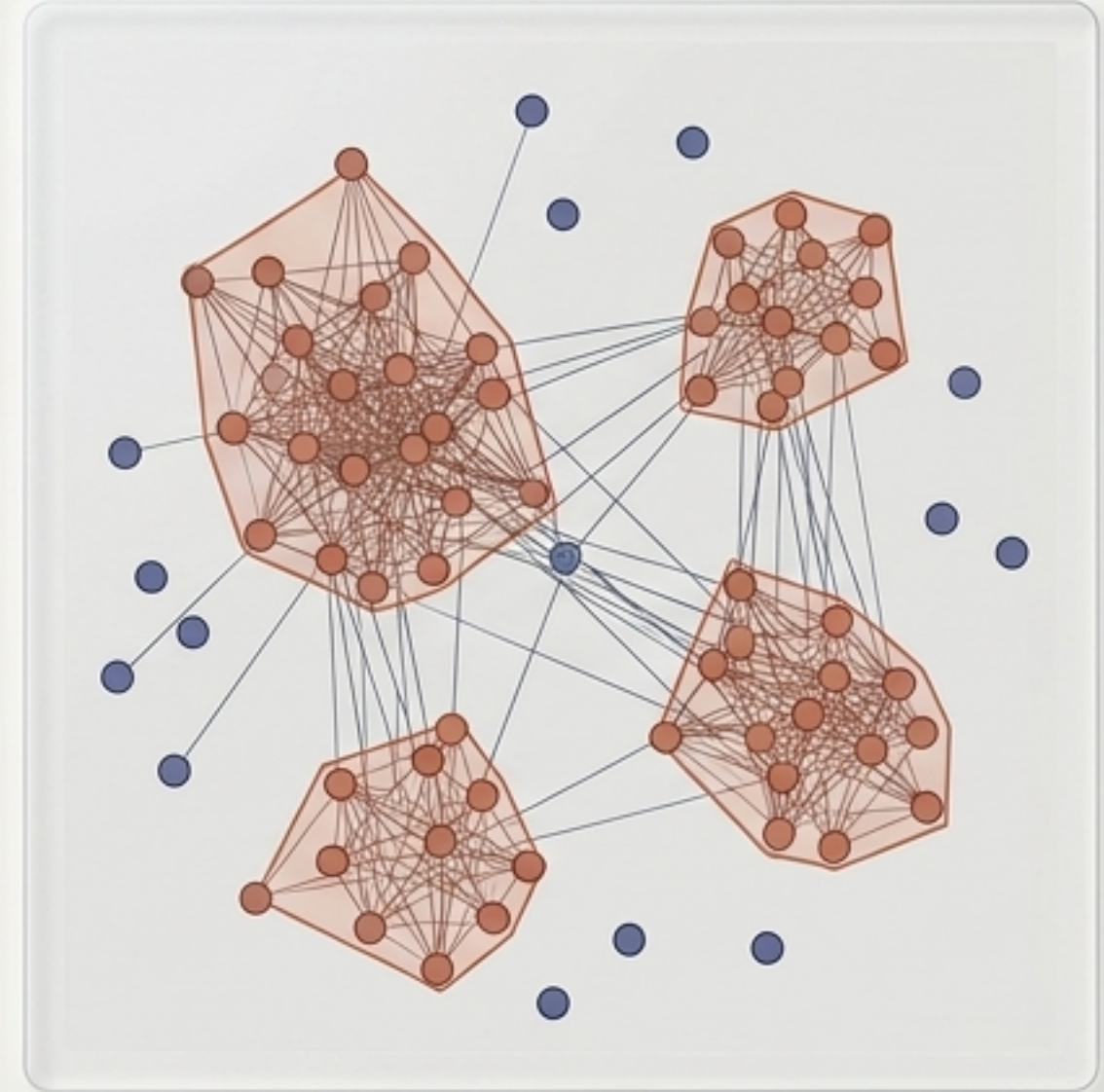
Edges are drawn if composite similarity $S_{\text{comp}} \geq 0.78$ across ≥ 3 layers.

2. Algorithm Process



The Newman-Girvan Modularity equation overlays the dynamic adjacency matrix.

3. Precision Output



Exact community boundaries mapped.

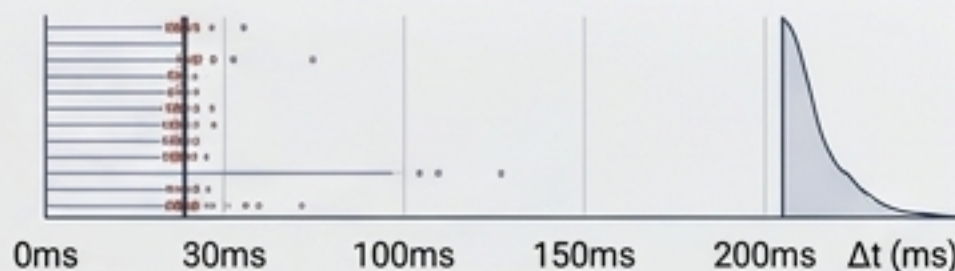
No need to guess the cluster size. Any community with $Q > 0.60$ and node count ≥ 3 is mathematically classified as a coordinated fraud syndicate ($P < 10^{-5}$).

Plain-English evidence replaces the black-box risk score

WHY CARD: Syndicate Cluster #01

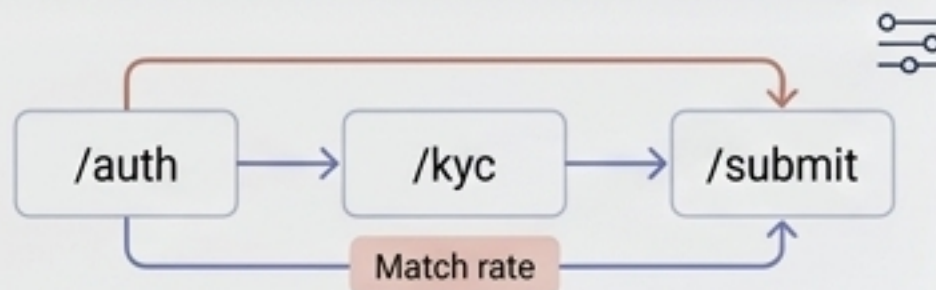
✓ Temporal Synchronization:

Average inter-arrival delta $\Delta t = 38\text{ms}$ across 20 accounts (Z-score = 4.82, $P < 0.0001$).



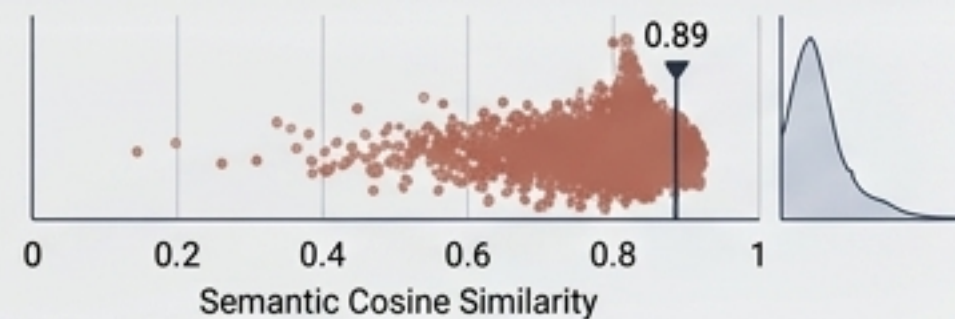
✓ Navigation Determinism:

96% Finite State Machine route identity (Sequence: /auth $\rightarrow$ /kyc $\rightarrow$ /submit).



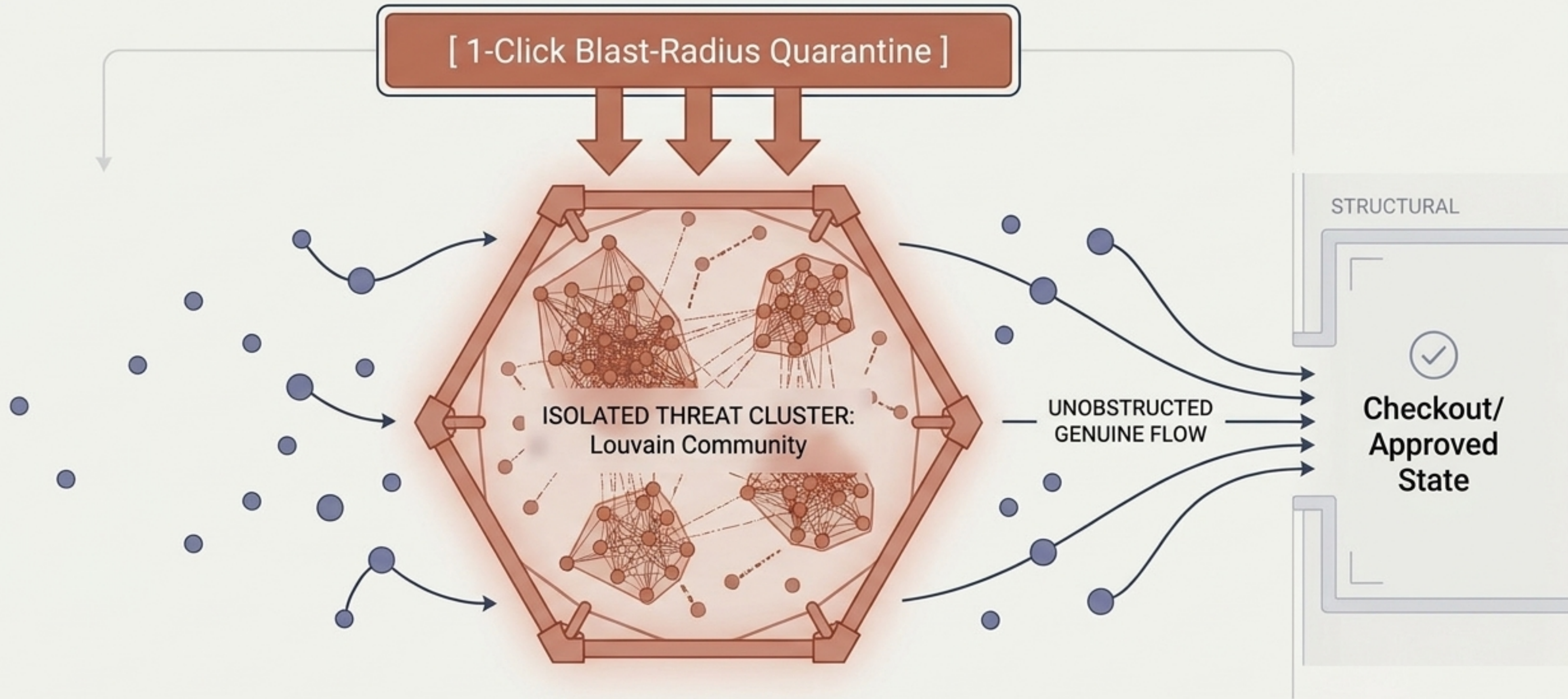
✓ Semantic Cosine Overlap:

Mean semantic vector similarity of 0.89 across 20 distinct loan bios.



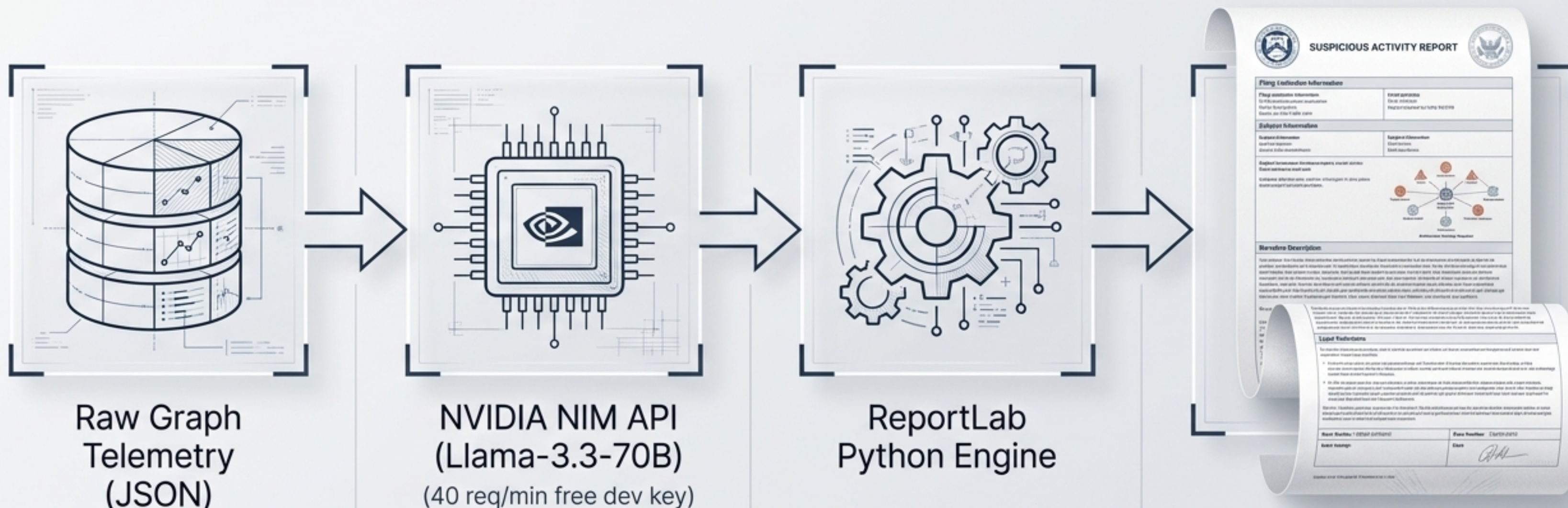
Zero Black-Box Mystery. Directly answers the CFPB mandate with plain-English adverse action justifications.

Fumigating the entire nest without collateral damage to genuine customers



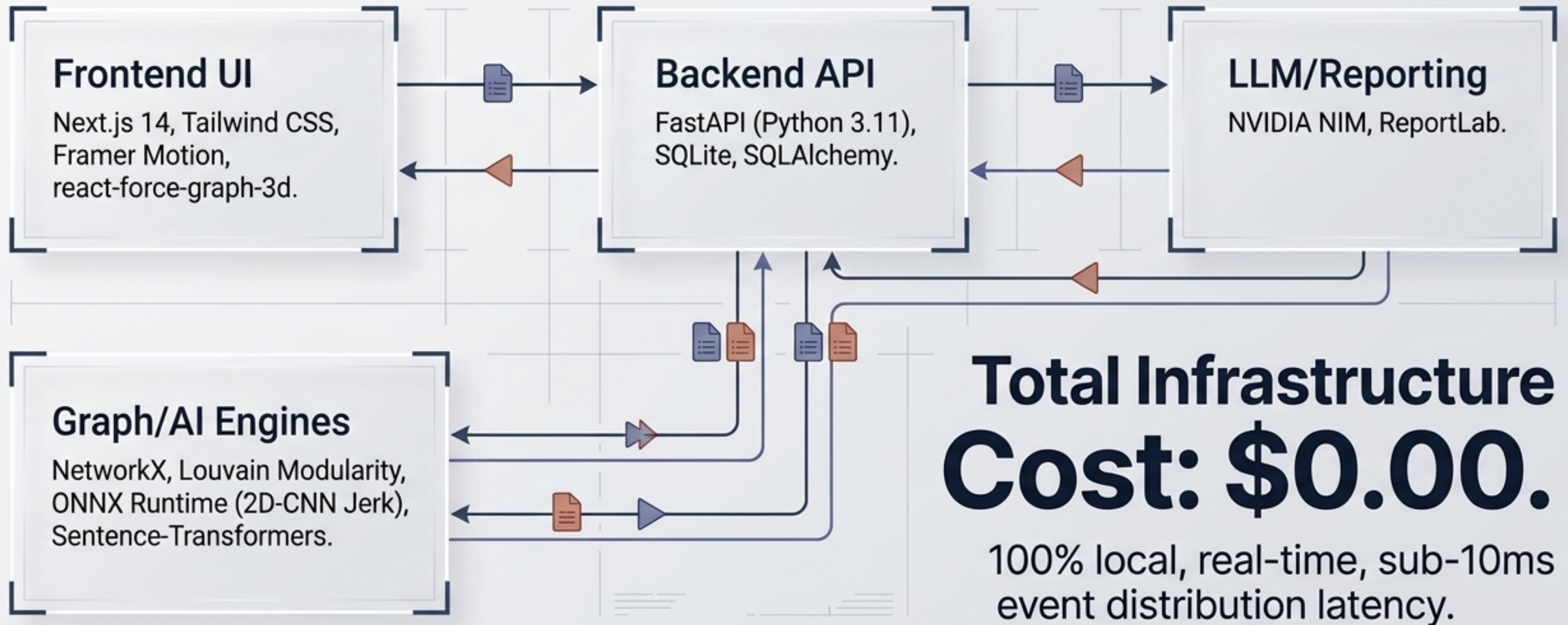
Legacy teams play "whack-a-mole" blocking one bot at a time. ShadowGram isolates the entire Louvain community instantly. Genuine customers experience zero friction, zero step-up challenges, and zero delays.

Raw graph telemetry becomes courtroom-ready regulatory documents in two seconds



Llama-3.3-70B acts as the legal scribe, converting JSON metrics into formal regulatory narratives complying with CFPB Circular 2023-03 and ECOA Regulation B.

A fully local, real-time cyber-forensics platform with zero cloud infrastructure bills



Protecting users, respecting privacy, and eliminating wasted verification costs



ShadowGram doesn't just catch scammers—it respects privacy, removes friction, and serves as the ultimate financial cyber-defense moat.